

August 23, 2026

Editor-in-Chief
Engineering Applications of Artificial Intelligence

Dear Editor-in-Chief,

Please find enclosed our manuscript entitled "Geodesic Diagnostic Trajectories (GDT): A Differential Geometry Framework for Adaptive Decision Support and Operational Optimization in EHR Systems" for consideration for publication in Engineering Applications of Artificial Intelligence. The work introduces a Riemannian manifold representation of evolving diagnostic intent and an associated curvature-gated retrieval strategy (GDT), and reports a reproducible, simulation-based validation and operational analysis connecting retrieval gains to queueing-model improvements in hospital throughput.

We confirm that this manuscript is original, not under consideration elsewhere, and that all authors have approved the manuscript and agree with its submission to Engineering Applications of Artificial Intelligence.

This manuscript is available as a preprint on SSRN.

Sincerely,

Hemanth Manchabale Papachappa
Aniruddha Ganguly